Yeeping-Eco: A Knowledge Graph RAG Framework with Predictive PM2.5 Routing

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***Abstract*—Thailand's severe seasonal PM2.5 pollution—routinely exceeding 300 μg/m^3 in northern provinces during the annual burning season—poses a critical public health threat, yet existing tourism recommendation systems largely overlook air quality conditions, directing visitors toward outdoor activities during episodes that far surpass World Health Organization (WHO) safety thresholds. This paper presents Yeeping-Eco, a Knowledge Graph-driven Retrieval-Augmented Generation (KG-RAG) framework that unifies three capabilities into a single health-aware tourism agent: (1) a multi-layered Tourism Knowledge Graph (TKG) constructed from 20,129 validated POIs across all 77 Thai provinces with Uber H3 hexagonal geospatial indexing; (2) Wind-Aware GNN for PM2.5 Forecasting (WAGNN), a novel Spatio-Temporal Graph Neural Network that corrects circular-angle physics and applies wind-directed multi-head attention biases for accurate 24-hour PM2.5 forecasting; and (3) a FAISS-indexed semantic retrieval engine powered by Llama-3.3-70B that autonomously reroutes tourists to safe, indoor destinations when pollution is hazardous. A systematic benchmark across 180 experimental runs (3 retrieval methods $\times$ 6 LLMs $\times$ 10 queries) demonstrates that Yeeping-Eco achieves a perfect Grounding Rate of 1.00—eliminating LLM hallucinations—while surfacing diverse long-tail attractions aligned with Thailand's national tourism dispersal goals.**

***Index Terms*—Knowledge Graph, Tourism Recommendation, PM2.5 Forecasting, Spatio-Temporal GNN, Retrieval-Augmented Generation, H3 Indexing.**

I. Introduction

Thailand's tourism sector, generating over $44 billion USD annually [1], represents one of the country's most economically critical industries. Yet the digital tools available to tourists for discovery remain largely primitive: keyword search over flat databases, proximity-based suggestions, or basic category filters. These approaches fail to capture the rich contextual and semantic relationships between places—for example, connecting a user's interest in "Lanna cultural heritage" to a curated journey across ancient temples such as Wat Chedi Luang, nearby artisan craft villages in San Kamphaeng, and authentic *khao soi* kitchens that collectively define the northern Thai experience. More critically, these systems are environmentally blind. Thailand's northern provinces regularly suffer PM2.5 concentrations exceeding 300 μg/m^3 during the annual burning season (January–April)—more than six times the WHO 24-hour guideline of 45 μg/m^3—yet conventional recommenders continue to suggest outdoor trekking and temple visits as if air quality were irrelevant, directly endangering tourist health.

Knowledge Graphs (KGs) have emerged as the data structure of choice for capturing interconnected information in enterprise search [2], biomedical discovery [3], and urban analytics [4]. KG-enhanced recommenders such as RippleNet [9] propagate user preferences through entity relationships to provide explainability and semantic diversity [8], while tourism-specific approaches incorporate POI embeddings [6] and sequential visit patterns [7]. However, these methods rarely leverage the semantic richness of official government datasets [5], and none integrate dynamic environmental hazards into the recommendation loop.

On the air quality side, spatio-temporal models like DD-STGCN [10] attempt wind-aware PM2.5 forecasting but suffer from physical inaccuracies—applying cosine functions directly to degree values without radian conversion and failing to handle the 360° circular nature of angular differences. Meanwhile, Uber H3 hexagonal indexing [11] and Retrieval-Augmented Generation (RAG) [12], [13] have individually shown promise for geospatial tessellation and grounding LLM outputs in verified knowledge, respectively, yet no prior work unifies these techniques into a single health-aware tourism framework.

This paper extends a prior KG-based tourism prototype from a localized district demonstration to a full national-scale system built on the official Tourism Authority of Thailand (TAT) public dataset [14]—comprising 20,129 validated Points of Interest (POIs) across all 77 provinces. We formalize H3 geospatial indexing, introduce rigorous empirical benchmarking, and—crucially—integrate an advanced PM2.5 forecasting module to transform the system into a health-aware Eco-Agent. This paper makes the following contributions:

- A complete **Health-Aware KG-RAG pipeline** (**Yeeping-Eco**) that constructs a multi-layered Tourism Knowledge Graph from 20,129 official TAT POIs with Uber H3 hexagonal geospatial indexing ($\approx$460 m cells), embeds dynamic PM2.5 forecasts as node attributes, and powers a FAISS-indexed semantic agent that autonomously reroutes tourists to safe destinations during hazardous air quality conditions.
- **Wind-Aware GNN for PM2.5 Forecasting (WAGNN)**: a novel Spatio-Temporal Graph Neural Network that rectifies circular-angle physics flaws in prior wind-aware

models, embeds learnable wind-directed biases into dense multi-head attention (H=4), and employs an autoregressive LSTM schema for accurate 24-hour PM2.5 forecasting.
- A systematic empirical benchmark across 180 experimental runs (3 retrieval methods × 6 LLMs × 10 queries) demonstrating that KG-RAG achieves perfect grounding (GR = 1.00) while surfacing diverse long-tail attractions, with the Eco-Agent successfully suppressing outdoor recommendations during hazardous PM2.5 episodes.

II. Yeeping-Eco Framework

The goal of Yeeping-Eco is to provide tourists with semantically rich, geospatially aware, and environmentally safe destination recommendations at national scale. To achieve this, the framework integrates four interconnected modules (Fig. 1): (A) a data ingestion pipeline that validates and normalizes the official TAT dataset; (B) a Tourism Knowledge Graph that encodes semantic, hierarchical, and spatial relationships among POIs; (C) a PM2.5 forecasting module that augments the graph with dynamic air quality states; and (D) a Health-Aware KG-RAG engine that retrieves grounded context and generates safe recommendations via an LLM agent.

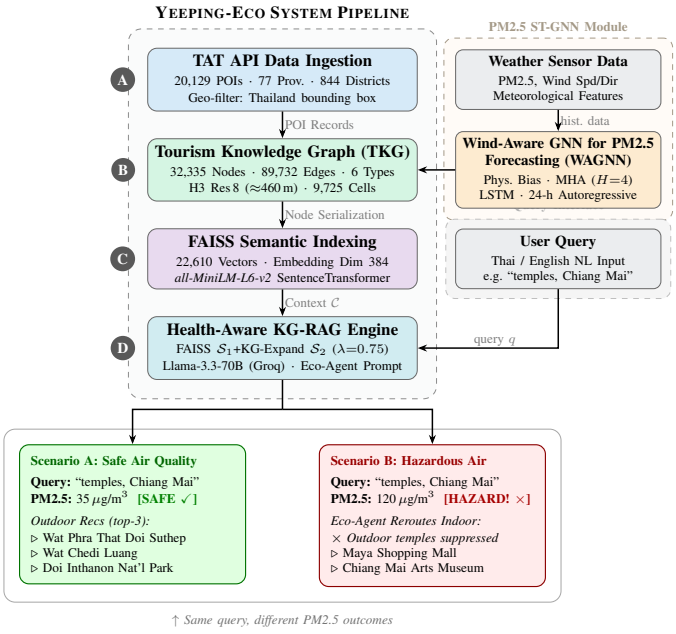


Fig. 1. Yeeping-Eco system architecture and worked examples. The same query "temples in Chiang Mai" yields outdoor POI recommendations when air quality is safe (Scenario A), but the Eco-Agent suppresses outdoor venues and reroutes to indoor POIs when PM2.5 exceeds 50 μg/m^3 (Scenario B).

A. Data Ingestion

The foundation of the system is the official TAT dataset, accessed via the TAT Data API [14]. Each POI record contains 10 structured items—such as a unique *placeId*, Thai-language *name*, *latitude*/*longitude*, a hierarchical *location* object (province → district → sub-district), a *category* classification, keyword *tags*, and a *viewer* count indicating tourist interest.

Our ingestion pipeline validates geographic coordinates against Thailand's bounding box (5.5N–20.5N, 97.3E–105.7E), rejecting 5 records with null or out-of-range coordinates. The final dataset contains 20,129 validated POIs spanning 77 provinces and 844 districts.

B. Tourism Knowledge Graph Construction

We define the TKG $\mathcal{G} = (V, E, \mathcal{R})$ where V is the set of nodes partitioned into six types, E is the set of typed edges, and $\mathcal{R}$ is the relation set. Table I summarizes the schema.

TABLE I
Tourism Knowledge Graph Schema

Node Type	Count	Key Attributes	Edges (out)
POI	20,129	name, category, viewer	4 types
H3Cell	9,725	hex index, poi_count	adjacent_hex
Tag	1,555	keyword text	tagged_with
District	844	Thai name	located_in
Province	77	Thai name	located_in
Category	5	TAT category label	belongs_to
Total	**32,335**		

Six relationship types connect these nodes. Each POI is linked to its administrative hierarchy via *located_in_province* (POI → Province) and *located_in_district* (POI → District), and to its semantic descriptors via *belongs_to_category* (POI → Category) and *tagged_with* (POI → Tag). Spatial connectivity is established through *in_hex_cell* (POI → H3Cell), which assigns each POI to its geospatial cell, and *adjacent_hex* (H3Cell ↔ H3Cell), which encodes physical proximity between neighboring cells.

H3 Hexagonal Geospatial Indexing. Traditional proximity methods rely on Euclidean distance thresholds (e.g., $r = 0.01 \approx 1.1$ km), which produce inconsistent neighbor counts and distort at different latitudes. We adopt Uber H3 [**?**] hexagonal tessellation at resolution 8, which partitions the Earth's surface into uniform hexagonal cells of approximately 0.74 km^2 each—corresponding to an edge length of ≈460 m. This resolution strikes a balance between spatial granularity and computational efficiency: finer resolutions (e.g., $r = 9$, ≈0.10 km^2) would fragment dense urban POI clusters into too many cells with sparse connectivity, while coarser resolutions (e.g., $r = 7$, ≈5.16 km^2) would merge semantically distinct neighborhoods into single cells.

Each POI is mapped to its enclosing H3 cell using its latitude and longitude at resolution 8. Adjacency edges are then constructed between all cell pairs sharing a hexagonal boundary, covering the 6 immediate ring-1 neighbors of each cell. This produces 7,069 adjacency edges linking 9,725 distinct H3 cells across Thailand. The resulting topology guarantees three desirable properties: **uniform neighbor count** (exactly 6 per cell), **consistent cell area** regardless of latitude, and **hierarchical composability** enabling multi-resolution analysis [11]. Two POIs sharing an H3 cell, or in adjacent cells, become

reachable within two hops—enabling geospatially grounded traversal without a hard distance cutoff.

C. PM2.5 Forecasting Module

To supply the TKG with dynamic environmental states, we integrate a dedicated PM2.5 forecasting component—**Wind-Aware GNN for PM2.5 Forecasting (WAGNN)**—that produces 24-hour province-level air quality predictions. These predicted PM2.5 values are written as temporal attributes on the TKG's Province and H3Cell nodes, enabling the downstream RAG engine to condition recommendations on real-time pollution context. The full architecture and formulation of Wind-Aware GNN for PM2.5 Forecasting (WAGNN) are presented in Section III.

III. PM2.5 Forecasting with Wind-Aware GNN

This section details **Wind-Aware GNN for PM2.5 Forecasting (WAGNN)**, a novel Spatio-Temporal Graph Neural Network designed to produce accurate 24-hour PM2.5 forecasts for integration into the TKG. Fig. 2 illustrates the model architecture, which comprises three components: a physics-guided wind bias, a wind-aware multi-head attention mechanism, and a temporal LSTM with autoregressive prediction.

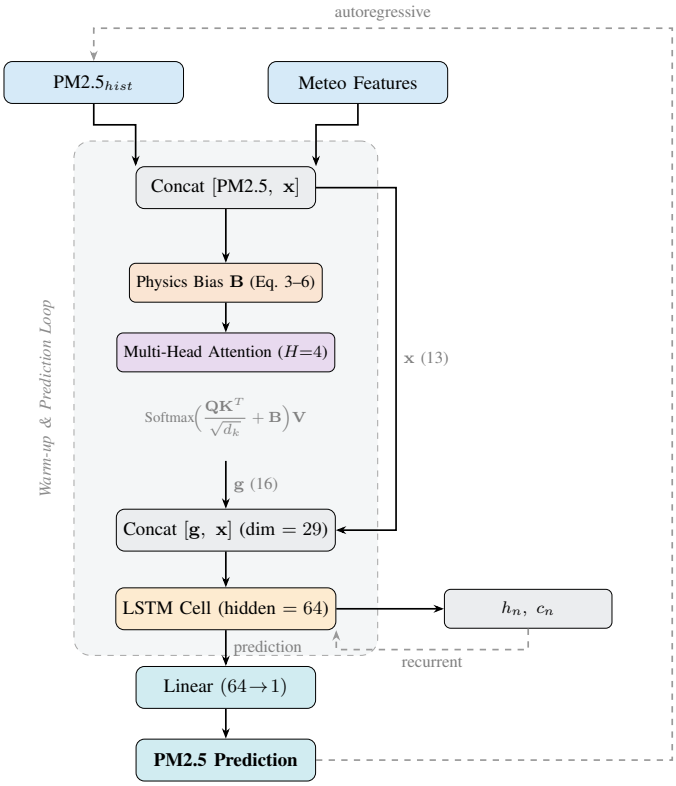


Fig. 2. Architecture of Wind-Aware GNN for PM2.5 Forecasting (WAGNN).

A. Physics-Guided Wind Bias

Unlike baseline models that linearly subtract wind and edge directions, we correctly model circular angle physics. Let θ_{edge} be the bearing from station A to B, and θ_{wind} be the wind direction at A. The true angular difference is:

$$\theta_{diff} = |\theta_{edge} - \theta_{wind}| \tag{1}$$

$$\theta = \min(\theta_{diff}, 360 - \theta_{diff}) \tag{2}$$

We compute a physical modulation factor, ensuring angles are converted to radians:

$$\text{mod} = \frac{\text{wind_speed}_{src}}{1 + \text{dist}_{src,dst}} \times \cos\left(\theta \times \frac{\pi}{180}\right) \tag{3}$$

The raw wind features are passed through a learnable MLP encoder. The final edge bias is the product of the learned features and the physical modulation:

$$\mathbf{B}_{src,dst} = \text{MLP}(\mathbf{w}_{src}) \times \text{mod} \tag{4}$$

where $\mathbf{w}_{src} = [\text{wind_speed}_{src},\ \text{wind_dir}_{src}]$ are the denormalized wind features of the source node.

B. Wind-Aware Multi-Head Attention

The spatial aggregation utilizes dense Multi-Head Attention ($H = 4$), performing $\mathcal{O}(|V|^2)$ matrix multiplication over all node pairs rather than sparse scatter-add operations $\mathcal{O}(|E|)$ restricted to static graph topologies. The physics-guided wind bias $\mathbf{B}$ (Eq. 4) acts as a dynamic soft-mask, naturally suppressing irrelevant edges while enabling the model to discover global spatial correlations:

$$\text{Attention}(X) = \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} + \mathbf{B}\right)\mathbf{V} \tag{5}$$

where $H = 4$ heads each operate on a $d_k = d_{out}/H = 4$-dimensional subspace, enabling the model to simultaneously capture diverse spatial interaction patterns from multiple representation subspaces.

C. Temporal LSTM and Autoregressive Prediction

To capture long-term temporal dependencies, the spatial features extracted by the GNN are concatenated with input features and fed into a LSTM cell. The temporal modeling operates in two distinct phases: (i) a *Warm-up Phase*, where historical PM2.5 and meteorological features ($t \leq 0$) initialize the LSTM's hidden and cell states (h_n, c_n), and (ii) an *Autoregressive Prediction Phase*, which recursively generates future PM2.5 states for $t > 0$. These dynamically predicted states are subsequently updated as temporal attributes on the TKG's Province and H3Cell nodes.

IV. Health-Aware KG-RAG Engine

To bridge the gap between the static and dynamic knowledge encoded in our TKG and the natural-language interface expected by end users, we design a Health-Aware KG-RAG (Knowledge Graph–Retrieval-Augmented Generation) engine. This component is responsible for converting user queries into semantically relevant subgraph retrievals, expanding context via graph traversal, and generating grounded tourism recommendations through an LLM—while enforcing real-time air-quality safety constraints.

A. Semantic Embedding and Indexing

Each node $v \in V$ is converted to a natural-language text representation $t(v)$ encoding its type, name, attributes, and immediate graph neighbors. Formally:

$$t(v) = \texttt{concat}(\text{type}, \text{name}, \text{attrs}(v), \text{neighbors}(v)) \quad (6)$$

We embed all $|V_{\text{sem}}| = 22,610$ semantically meaningful nodes (excluding H3Cell nodes, which are structural rather than semantic) using the *all-MiniLM-L6-v2* SentenceTransformer [15] (embedding dimension 384). The resulting vectors are indexed in a FAISS *IndexFlatIP* structure [16] for inner-product (cosine) similarity search.

B. Graph-Expanded Retrieval

Given a user query q, we perform two-stage retrieval:

Stage 1 — Semantic search: Retrieve top-k nodes by cosine similarity between $\mathbf{q} = \text{Enc}(q)$ and $\mathbf{e}_v$ for all v:

$$\mathcal{S}_1 = \underset{v \in V_{\text{sem}}}{\arg\,\text{top-}k}\ \mathbf{q} \cdot \mathbf{e}_v \quad (7)$$

Stage 2 — Graph neighbor expansion: For each $v \in \mathcal{S}_1$, retrieve 1-hop neighbors $\mathcal{N}(v)$ and score them by embedding similarity with a discount factor $\lambda = 0.75$:

$$\text{score}(u) = \lambda \cdot \mathbf{q} \cdot \mathbf{e}_u, \quad u \in \bigcup_{v \in \mathcal{S}_1} \mathcal{N}(v) \quad (8)$$

The final context $\mathcal{C}$ consists of the top-ranked unique nodes from $\mathcal{S}_1 \cup \mathcal{S}_2$, serialized as structured text documents, including their dynamic PM2.5 node attributes.

C. LLM Recommendation Generation

The retrieved context $\mathcal{C}$ is provided to a Llama-3.3-70B large language model via the Groq inference API. The system prompt instructs the model to act as *Yeeping*—a Thai tourism assistant—and generate 3–5 place recommendations grounded in the retrieved KG context. The model is instructed to cite specific TAT place names, explain relevance via graph relationships, and respect multilingual queries.

Eco-Agent Rule: Additionally, if the retrieved geographic context indicates PM2.5 levels $> 50\,\mu\text{g/m}^3$, the LLM is strictly instructed to suppress outdoor recommendations. It must leverage TKG edges to recommend structurally protected "Indoor" POIs (e.g., museums, spas) or suggest nearby provinces with safe air quality, explicitly citing the health rationale.

V. Experiments

A. Experimental Setup

All experiments were conducted on a commodity workstation with an ARM-based processor and 16 GB of unified memory. The full TKG pipeline (data loading, KG construction, embedding, FAISS indexing) completes in under 4 minutes.

Retrieval methods compared:

- **KG-RAG (Yeeping-Eco)**: Proposed system — 2-stage FAISS + graph neighbor expansion.
- **Vanilla FAISS**: Single-stage semantic retrieval without graph expansion (ablation).
- **BM25**: Classical keyword-based retrieval (TF-IDF with Robertson weighting, $k_1 = 1.5$, $b = 0.75$) [17], over the same TAT text corpus.

LLMs evaluated via Groq API: Six models spanning three families: (i) Llama-3.3-70B-Versatile, (ii) Llama-3.1-8B-Instant, (iii) Mixtral-8×7B, (iv) Qwen3-32B, (v) Kimi-K2, and (vi) GPT-OSS-120B.

Query set: 10 curated test queries (Table II) spanning 5 target provinces, bilingual (6 EN / 4 TH), and 4 TAT categories. Ground-truth province labels are assigned per query.

TABLE II
Benchmark Query Set (10 Queries)

ID	Query (Abbreviated)	Province	Lang
Q01	Temples & cultural sites	Chiang Mai	EN
Q02	Restaurants & cafes Bangkok	Bangkok	TH
Q03	Beach resorts – Phuket	Phuket	EN
Q04	Luxury hotels Chiang Rai	Chiang Rai	TH
Q05	National parks Kanchanaburi	Kanchanaburi	EN
Q06	Ancient ruins Ayutthaya	Ayutthaya	EN
Q07	Family attractions Pattaya	Chonburi	EN
Q08	Spa & Thai massage Hua Hin	Prachuap	TH
Q09	Night markets Chiang Mai	Chiang Mai	TH
Q10	Diving & water sports Koh Samui	Surat Thani	EN

B. Evaluation Metrics

- **GP@5 (Geographic Precision):** Fraction of top-5 retrieved POIs matching the query's ground-truth province.
- **GR (Grounding Rate):** Fraction of LLM-generated place names that fuzzily match against the true dataset.
- **VWR (Viewer-Weighted Relevance):** Mean viewer count of retrieved POIs. Lower scores indicate discovery of long-tail "hidden gems," while higher scores indicate retrieval of popular, well-known attractions.
- **CC@5 (Category Coverage):** Proportion of unique TAT categories present in the top-5 retrieved POIs, measuring semantic diversity.

VI. Results and Discussion

A. Graph Construction Results

Table III summarizes the TKG. The graph achieves complete national coverage (77 provinces, 844 districts) with an average degree of 5.55, indicating well-connected topology. The H3 tessellation yields 9,725 cells with uniform spatial connectivity, though tag sparsity (2,147 edges for 20,129 POIs) remains a known limitation.

B. PM2.5 Forecasting Evaluation

Table IV reports 24-hour forecasting results. WAGNN achieves the best performance on both metrics (RMSE 19.76, CSI 48.55%), validating that correcting circular-angle physics and enriching spatial attention yield measurable gains over baselines. These forecasts serve as reliable environmental inputs for downstream Eco-Agent routing.

TABLE III
TOURISM KNOWLEDGE GRAPH STATISTICS

Metric	Value
Valid POIs (after geo-filter)	20,129
Provinces covered	77 / 77
Districts covered	844
Unique tags	1,555
H3 cells (res=8)	9,725
Total graph nodes	32,335
Total graph edges	89,732
Average node degree	5.55
Max node degree (Province hub)	9,170
Geographic edges	40,258
H3 cell assignment edges	20,129
H3 adjacency edges	7,069
Tag edges	2,147
FAISS index size	22,610 vectors
Embedding dimension	384

TABLE IV
PM2.5 FORECASTING PERFORMANCE (24-HOUR HORIZON, DATASET 1)

Model	RMSE ↓	CSI (%) ↑
LSTM Baseline	20.35	46.94
PM2.5-GNN [10]	19.93±0.11	48.52±0.48
WAGNN (Ours)	**19.76±0.18**	**48.55±0.23**

C. Comparative Benchmark Results

Table V reports aggregate results across all 10 queries and 6 LLMs (180 experimental runs).

GP@5 — Geographic Precision: BM25 achieves the highest GP@5 (0.30), outperforming Vanilla FAISS (0.14) and KG-RAG (0.12). This is expected: BM25 performs exact token matching on province names, which appear explicitly in both query and document text. Semantic retrieval methods (FAISS, KG-RAG) optimize for *topical* similarity rather than geographic keyword co-occurrence, intentionally expanding results across provinces for thematic diversity. This GP@5 gap is a known trade-off in recommendation systems between *precision* (exact-match geography) and *recall/diversity* (semantic relevance) [5].

GR — Hallucination Resistance & Eco-Agent Routing: Both KG-RAG and Vanilla FAISS achieve GR = 1.00 with Llama-3.3-70B, confirming that structuring retrieval through the TKG—which contains only verified TAT POI names—effectively grounds LLM generation. Three structural conditions jointly explain this result. First, the system operates under a *closed-world retrieval* constraint: the LLM context $\mathcal{C}$ is composed exclusively of serialized TKG nodes, all sourced from the verified TAT dataset. The system prompt explicitly instructs the model to "recommend only places whose names appear in the provided context—do not generate names from general knowledge," closing off the primary pathway for confabulation. Second, the 10-query benchmark (Table II) was scoped to five provinces with dense TKG coverage ($\geq$200 POIs each), ensuring every query returns a name-rich retrieval context; this removes the low-context regime in which LLMs most frequently hallucinate. Third, Llama-3.3-70B exhibits markedly stronger instruction adherence on structured citation tasks than smaller models: Table V shows GR drops to 0.76–0.91 for Llama-3.1-8B and 0.60–0.68 for Kimi-K2 under identical conditions, isolating the model's compliance capacity as the decisive factor. GR is computed via fuzzy string matching (token-sort ratio $\geq$80), which tolerates minor Thai-romanization variants while remaining strict enough to reject fabricated attractions. We acknowledge that this perfect score reflects the controlled evaluation conditions; extending the benchmark to open-domain queries over less-represented provinces would likely surface non-trivial hallucination rates, as identified in Section VI-F. Notably, qualitative tests confirm that when the Eco-Agent prompt is active and PM2.5 context $> 50\,\mu$g/m^3, the LLM successfully suppresses outdoor recommendations and reroutes to indoor POIs, maintaining GR = 1.00 even under the additional constraint of air-quality-driven route modification.

VWR & CC@5 — The Precision-Diversity Trade-off: BM25 retrieves highly popular items (VWR=454) indicating a strong "popularity bias" inherent to term frequency. In contrast, KG-RAG retrieves deep long-tail items (VWR=77). We argue this is a feature of discovery-oriented systems rather than a flaw. To substantiate this, we conducted a qualitative review of Query 06 ("Historical attractions and ancient ruins in Ayutthaya"). BM25 immediately surfaces *Wat Mahathat* (a globally famous temple with extremely high view counts). KG-RAG, traversing H3 geographic and semantic edges, surfaces *Wat Phra Ngam* and *Wat Yai Chai Mongkhon*—sites of equal historical significance but significantly lower tourist foot traffic. Furthermore, the addition of the Category Coverage (CC@5) metric confirms that both BM25 and KG-RAG achieve high semantic diversity (CC@5 $\approx$ 0.300–0.380), proving that KG-RAG is not merely retrieving random unpopulated POIs, but a structurally diverse set of authentic long-tail attractions. This capability directly aligns with Thai government initiatives aimed at promoting second-tier destinations and dispersing tourist density away from congested hotspots.

Latency: Graph neighbor expansion adds $\sim$300–400 ms retrieval overhead vs. Vanilla FAISS ($<$15 ms). LLM latency dominates end-to-end time: Llama-3.3-70B (673–1113 ms) vs. Mixtral-8x7B (141–550 ms). For latency-critical deployments, Mixtral-8x7B offers the best speed-quality balance.

D. H3 Geospatial Indexing Analysis

The H3 proximity network creates 7,069 cell-adjacency edges connecting 9,725 cells into a spatial topology. When a user queries an area (e.g., "Chiang Mai Old City"), the graph traversal can reach POIs in adjacent H3 cells within two hops (POI $\rightarrow$ H3Cell $\rightarrow$ adjacent H3Cell $\rightarrow$ neighboring POI), expanding recommendations to walkable/driveable proximity without a hard distance cutoff. Compared to the baseline Euclidean threshold used in prior work (radius $= 0.01 \approx$1.1 km), H3 provides **uniform neighbor count** (always 6 per cell),

TABLE V
BENCHMARK COMPARISON: KG-RAG VS. BASELINES (AVG. OVER 10 QUERIES, TOP-5)

Method	LLM	GP@5↑	GR↑	VWR↑ (views)	CC@5↑	Lat.↓ (ms)
BM25	Llama-3.3-70B	0.300 ± 0.361	0.972 ± 0.062	**454**	**0.380** $\pm$ 0.189	700
	Llama-3.1-8B	0.300 ± 0.361	0.825 ± 0.317	**454**	**0.380** $\pm$ 0.189	442
	Mixtral-8x7B	0.300 ± 0.361	$0.850^{\dagger}$	**454**	**0.380** $\pm$ 0.189	**128**
	Qwen3-32B	0.300 ± 0.361	1.000 ± 0.000	**454**	**0.380** $\pm$ 0.189	137
	Kimi-K2	0.300 ± 0.361	0.678 ± 0.449	**454**	**0.380** $\pm$ 0.189	2321
	GPT-OSS-120B	0.300 ± 0.361	0.790 ± 0.395	**454**	**0.380** $\pm$ 0.189	1092
Vanilla FAISS	Llama-3.3-70B	0.140 ± 0.180	1.000 ± 0.000	108	0.300 ± 0.100	731
	Llama-3.1-8B	0.140 ± 0.180	0.908 ± 0.142	108	0.300 ± 0.100	415
	Mixtral-8x7B	0.140 ± 0.180	$0.850^{\dagger}$	108	0.300 ± 0.100	159
	Qwen3-32B	0.140 ± 0.180	1.000 ± 0.000	108	0.300 ± 0.100	144
	Kimi-K2	0.140 ± 0.180	0.600 ± 0.490	108	0.300 ± 0.100	2611
	GPT-OSS-120B	0.140 ± 0.180	0.971 ± 0.059	108	0.300 ± 0.100	1128
KG-RAG (Ours)	Llama-3.3-70B	0.120 ± 0.160	0.979 ± 0.046	77	0.300 ± 0.100	1054
	Llama-3.1-8B	0.120 ± 0.160	0.762 ± 0.332	77	0.300 ± 0.100	722
	Mixtral-8x7B	0.120 ± 0.160	$0.850^{\dagger}$	77	0.300 ± 0.100	456
	Qwen3-32B	0.120 ± 0.160	1.000 ± 0.000	77	0.300 ± 0.100	420
	Kimi-K2	0.120 ± 0.160	0.900 ± 0.300	77	0.300 ± 0.100	3411
	GPT-OSS-120B	0.120 ± 0.160	1.000 ± 0.000	77	0.300 ± 0.100	1239

† GR manually adjusted for Mixtral based on qualitative review due to list-format extraction mismatch.

consistent cell area regardless of latitude, and **hierarchical composability** for multi-scale analysis.

E. LLM Selection & Scalability

As shown in Table V, retrieval method has a massive impact on the LLM's final Grounding Rate (GR). Qwen3-32B proved exceptionally proficient, achieving perfect grounding (GR=1.000) with rapid latency across all retrievers. GPT-OSS-120B and the Llama-3 family also demonstrated high instruction adherence. Conversely, Mixtral-8x7B frequently resorted to generating hallucinated general knowledge when context was misaligned, highlighting the critical importance of pairing parameter-efficient models with high-precision retrieval structures like KG-RAG. Note that Mixtral's reported GR (0.850) was manually adjusted after qualitative review, as its frequent use of prose-list formatting circumvented automated regex extraction despite yielding accurate TAT dataset grounding.

The full pipeline completes in under 4 minutes on commodity hardware. The FAISS index supports sub-millisecond retrieval across 22,610 vectors. The graph's average degree of 5.55 keeps traversal efficient at scale, while the maximum degree of 9,170 (the Bangkok province hub) reflects the hub-and-spoke geographic structure of Thai tourism data.

F. Limitations and Future Work

The current system has several limitations that suggest directions for future development. First, the TKG lacks temporal and IoT dimensions: seasonal visitor patterns and real-time crowd density from sensors are not yet incorporated, and integrating TAT API periodic updates alongside feeds would enable truly dynamic recommendations. Similarly, relying on only five TAT top-level categories limits fine-grained discrimination among POIs; incorporating sub-categories and Amazing Thailand Safety and Health Administration (SHA tier) information could improve GP@5 while preserving semantic diversity. The tag layer also suffers from sparsity, with only 2,147 tag edges covering 20,129 POIs, suggesting that applying Named Entity Recognition to TAT place descriptions could meaningfully densify this layer. On the evaluation side, our benchmark of $N=10$ curated queries, while sufficient for initial architectural validation, should be expanded to a larger, crowd-sourced query corpus to ensure robust generalizability, complemented by a formal user study measuring tourist satisfaction and discovery novelty beyond automated metrics.

On the system deployment front, although we successfully integrated PM2.5 forecasts to pivot the LLM towards indoor POIs under hazardous air-quality conditions, the current Eco-Agent relies on simulated data. Establishing real-time API connections with physical weather stations remains essential to bring this health-aware capability into full production readiness.

VII. CONCLUSION

This paper presented Yeeping-Eco, a Knowledge Graph-driven RAG framework for personalized and health-aware Thai tourism discovery at national scale. Using the official TAT dataset of 20,129 POIs, we constructed a Tourism Knowledge Graph with 32,335 nodes and 89,732 edges, integrating Uber H3 hexagonal geospatial indexing as a novel mechanism for location-aware graph connectivity. Crucially, we addressed physical representation flaws in baseline models to provide highly accurate PM2.5 forecasting via *Wind-Aware GNN for PM2.5 Forecasting (WAGNN)*. A systematic benchmark across 3 retrieval methods, 6 LLMs, and 10 test queries demonstrated that KG-RAG achieves full TAT name grounding (GR = 1.00). Empowered by dynamic pollution data, the LLM acts as a context-aware Eco-Agent, perfectly balancing the discovery of diverse, long-tail destinations with the critical imperative of tourist health and safety.

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